

Columnar Transposition Cipher

Aim

To implement the keyword columnar transposition cipher. The plaintext is written row by row into a grid whose width equals the key length; the columns are then read out in the order given by ranking the key's letters alphabetically (ties keep their left-to-right order). It only reorders characters, so spaces, punctuation and case are preserved.

Algorithm

```
1 Function ColumnOrder(key):
2   return column indices sorted by key letter (stable, ascending)
3 Function Encrypt(plain_text, key):
4   cols ← length of key
5   order ← ColumnOrder(key)
6   cipher ← ""
7   foreach col in order do
8     idx ← col
9     while idx < length of plain_text do
10      append plain_text[idx] to cipher
11      idx ← idx + cols                                // step down the column
12 return cipher
```

```
1 Function Decrypt(cipher_text, key):
2   cols ← length of key
3   len ← length of cipher_text
4   order ← ColumnOrder(key)
5   fullRows ← len div cols
6   remainder ← len mod cols
7   foreach col in order do
8     height ← fullRows + (1 if col < remainder else 0)
9     column[col] ← next height characters of cipher_text
10  plain ← ""
11  for i ← 0 to len - 1 do
12    c ← i mod cols
13    append next unread character of column[c] to plain    // read row by row
14 return plain
```

Output

Encrypt — key ZEBRA, plaintext MEETATNOON

The key ranks its columns alphabetically (A is lowest); ties would keep left-to-right order. Below each column header is its read rank:

Column	Z	E	B	R	A
read rank	4	2	1	3	0

The plaintext is written row by row into a 5-column grid:

Z E B R A
M E E T A
T N O O N

Columns are then read out lowest rank first (A, B, E, R, Z):

Read order	Column	Rank	Characters
1	A	0	A N
2	B	1	E O
3	E	2	E N
4	R	3	T O
5	Z	4	M T

Result: MEETATNOON -> ANEOENTOMT

Decrypt — key ZEBRA, ciphertext ANEOENTOMT

With 10 characters over 5 columns there are 2 full rows and no remainder, so each column holds exactly 2 characters. The ciphertext is sliced back in read order:

Read order	Column	Height	Slice
1	A	2	AN
2	B	2	EO
3	E	2	EN
4	R	2	TO
5	Z	2	MT

Reading the reconstructed grid row by row (Z E B R A per row) recovers the text.

Result: ANEOENTOMT -> MEETATNOON